STA 220 Final Report

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Code

https://github.com/MaxwellCapron/STA220-NBA-Project/tree/FinalSubmission

Abstract

The NBA is one of the world's most competitive professional sports leagues, where teams are constantly changing their strategy to gain some sort of an edge on their opponents. There are also a plethora of statistics that are tracked, generating a ton of data for anyone who wants to perform analyses. For this project, we decided to use the NBA-API to pull player and in-game shot location data to analyze and visualize some of these league trends, specifically with regards to three point shot frequency. We limited the scope of our study to the center position as it highlighted this trend, and it was easier to pull the data without getting flagged. After compiling the dataset, we ran a longitudinal analysis of the volume, efficiency, and expected point value of different shot types by zone and topped it off with a chronological heat map visualizing every center field goal attempt from the regular season since 2013. We determined that there has been a notable uptick in centers shooting threes and also in their efficiency doing so. This has mostly come at the expense of mid-range jumpers, which are seen as inefficient due to a combination of their further distance from the hoop and lower payoff (2 points vs. 3) when compared with shots from behind the arc. Additionally, we found that the raw percentage increase of shots taken by centers from beyond the three point line was roughly equivalent to the increase among the rest of the league.

Introduction

Over the course of the past 10 years, the NBA has experienced what has been called a "Three Point Revolution." Stephen Curry, the undisputed greatest shooter of all time, and the title-winning 2014-2015 Golden State Warriors are widely considered to have been a catalyst for this leaguewide shift. We wanted to quantify this change in three point frequency across all teams. We also believe that this trend is most noticeable among players at the center position. NBA centers are traditionally the tallest and strongest players on the roster, which is typically paired with them being the least mobile. For many years, teams' offensive gameplans revolved around "posting up", which occurs when the center receives the ball in the midrange area and

backs their defender down to get as close as possible to the hoop for a high percentage shot. However, this strategy can easily be countered by having multiple defenders help, or by intentionally fouling (centers are typically not strong free throw shooters – see "Hack-a-Shaq.") Thus, the need for shooters to space the floor is paramount. Furthermore, teams began to realize that having centers who hit threes is the most effective way to open up space for teammates to attack the hoop as this drags the opposing center (usually the best interior defender) away from the basket. All this brings us to our main question of interest: how has NBA center shot selection and efficiency changed from 2013-2025? This is mainly an explorative study that will hopefully inspire further research questions as there are many avenues for potential analysis. As previously mentioned, we collected the data from nba.com using the NBA-API. The specific variables of interest were player id, player name, shot type, and shot location (consisting of specific X and Y coordinates that translate to the spot on the court from which the shot was taken.) The shot location portion of the dataset included 2,510,338 unique shot attempts for all player positions. The player info portion included over 700 total centers. One main assumption that was made was that players were strictly limited to their position labels. For example, we assume that any player listed as a center played all of their minutes as a center. In reality most players fill multiple positional roles depending on the situation, however this was a necessary assumption as it would be near impossible for the NBA to record in-game position changes in a sport that has been constantly moving away from traditional rigid positional roles. In addition, many players were labeled as hybrid positions such as "Forward-Center" however for the purposes of this study were decided to only look as the subset of players who were listed as "Center." Prior to our analysis, we believed the data would show a noticeable increase in three point frequency and this was largely confirmed. Even so, we were able to gain some additional insights that we will expand upon in our results section.

Methods

To begin, an examination of the NBA-API documentation was required to be able to develop a road map on how we were going to acquire all the necessary information to answer our questions of interest and visualize our results. We discovered that there were three necessary APIs to achieve our goals: Player Module, CommonPlayerInfo, and ShotChartDetail. The Player Module was utilized to create lists of all active and inactive NBA players. The CommonPlayerInfo endpoint was required to fetch necessary player information, the information we decided to extract were: person id, first and last name, position, height, and team name. Lastly, the ShotChartDetail endpoint allowed us to retrieve relevant shot information such as; player id, player name, team name, quarter, event type (converted shot or failed shot attempt), shot type (two-pointer or three-pointer), shot distance from basket, and the exact X/Y shot coordinates. Once the origin of our data was determined, we decided that we were to store two separate data frames, named "Centers" and "ShotLocs," into a single database named "nba_centers_info." Then, once stored properly, we were to utilize the Inner Join technique, matching on the

“ShotLocs” data frame player id column and the “Centers” data frame person id column, to create data tables for various testing and visualizations. Now that our project blueprint was established, the data acquisition began. With prior knowledge of the sensitivity the NBA-APIs obtained to a large load of requests in a small amount of time and also the amount of time it takes to acquire a large amount of data if we avoided being timed out, two separate functions were developed to acquire the active and inactive NBA centers. The reasoning was that we desired to get the code ready for data retrieval with a fraction of players before attempting to get the majority of it to avoid wasting time and making an excessive amount of requests. The acquisition of the 530 active players ran smoothly. On the other hand, issues arose when retrieving the data for the 4,573 inactive players. The particular issue was our code crashing due to being timed out for reaching the request limit of the CommonPlayerInfo API. Several strategies were attempted to bypass the rate limit issue, including increasing the delay in between requests and incorporating random intervals for our requests to simulate a more human-like behavior, unfortunately both remedies failed. Ultimately, the issue was resolved by manually requesting the information in batches of 100 players, along with an inconsistent wait period between requests. After all the issues were addressed, we filtered the data to only append the players whose position was listed as “Center” into a data frame called “Centers,” consisting of all 701 active and inactive centers. Next, another function was developed to create the second and last data frame. Due to inconsistent and incomplete data prior to the 2013-2014 NBA season, shot data from the 2013-2014 through 2024-2025 seasons was used to address our research questions. Additional filtering was required to refine our data frame, limiting the data to field goal attempts from regular season games. This excluded shot data, such as free throws and playoff game data from the analysis. Unexpectedly, this process was completed in a matter of minutes, a small fraction of the time it took the inactive centers function to complete, despite having 2,510,338 shot attempts by players for all positions. With both data frames successfully, it was necessary to combine them by taking the appropriate indexes from each. To achieve this, an inner join was performed on “ShotLocs” index player_id, matching to the “Centers” index person_id.

Once we performed this inner join, our final dataset was nearly complete. The only changes needed was to create a year variable by selecting the last four characters of the date string, and to assign every shot to a region of the court based on its X and Y location: above-the-break three, corner three, middle range, from the paint, and restricted area. Our final dataset had the following variables: PLAYER_NAME, POSITION, GAME_DATE, TEAM_NAME, EVENT_TYPE, SHOT_TYPE, SHOT_DISTANCE, X_LOC, Y_LOC, and SHOT_ZONE. The final dataset had 201,038 observations, there were no missing values as the NBA keeps solid records of their game stats. Now that our dataset is ready, we began visualizing the data.

In order to visualize the shot locations in each year, we created a gif with a heat map of the shot locations taken by centers in every year from 2013 to 2025. First, we created a function that would create an outline of a court underlying the data when creating our visualization in

matplotlib. Then, we divided the court into hexes (gridsize = 120) and assigned shots to those hexes by year. Our initial visualization wasn't useful, as due to the majority of shots by centers being taken in the restricted area, which is tiny relative to the other zones, the heat map was dominated by shots taken from this location. To rectify this, we used a logarithmic function to better visualize shots taken from the other zones. We then created line charts visualizing the percentage of shots taken beyond the three point line from 2013 to 2025, the percentage of shots taken from each zone over this period, and the expected value of shots taken per zone per year.

Results

Once we found all the shots taken by centers over the 2013-2025 period, we had the data needed to perform our analysis. The first thing we did was assigning every shot to a region of the court based on its X and Y location: above-the-break three, corner three, middle Range, from the paint, and restricted area, which allowed us to observe the change in shot patterns over the 12 year period.

Over the period of 2013 to 2025, we found a dramatic transformation in the frequency of three point shots among centers in the NBA. In 2013, only 1.4% of shots taken by centers were beyond the three point line, remaining relatively constant until 2016, when the number suddenly jumped to 4.9%. This continued increasing until a local peak in 2020 of 14.5%, dipping to 11.6% in 2023, and finally reaching another peak of 18.3% in 2025. This shift signals a fundamental redefinition of center play, where three point field goals have gone from exceptionally rare to over ⅙ of all shots taken by centers in recent years, or a 1200% increase. Interestingly, the increase in three point field goal attempts among centers mirrors the increase across the league. In 2013, 25.6% of field goal attempts among all shots were three point attempts, jumping to 41.9% in 2025, an increase of 16.3%. This is very similar to the 16.9% increase among centers over this period. In order to better understand the differences in three point field goal adoption between centers and the rest of the league, we will conduct log likelihood and OLS regression tests.

First, we conducted a log likelihood analysis on how the rate of change of the log odds differs between centers and the league. The following is the formula used in this analysis:

$$\log(p/(1-p))=\beta0+\beta1(\text{Year})+\beta2(\text{IsCenter})+\beta3(\text{Year}\times\text{IsCenter})$$

Where p is the probability a shot is taken beyond the three point line, β0 is the intercept, β1 is the change over year, β2 is added to the intercept if the player is a center, and β3 is the additional change over year if the player is a center. We are interested in testing the significance of β3, as if it is significant we can reject the hypothesis that the rate at which the log odds of shooting from beyond the three-point line changes by year is the same between centers and the overall league.

Running the test, we found that $\beta 3$ had a value of 0.098 and was statistically significant ($p \approx 0$) on top of the overall league's increase of 0.061 by year. This means centers adopted the three point shot at over twice the rate of the rest of the league.

The second test we ran in this trend comparison was an OLS regression on the difference in slope for the raw percentage rates. The following formula was used for this regression.

$$y=\beta 0+\beta 1(\text{Year})+\beta 2(\text{IsCenter})+\beta 3(\text{Year}\times\text{IsCenter})$$

Again, we are interested in $\beta 3$. The null hypothesis states that $\beta 3 = 0$, or that the annual percentage point change in the share of shots from beyond the three point line is the same between centers and the rest of the league. Running the test, we found $\beta 3$ had a coefficient of -0.0011, with a statistically insignificant p-value of 0.626. Therefore, we can determine that the annual change in percentage point in the share of shots beyond the three point line is the same between centers and the overall league. These two results paint an interesting picture. Centers started from a much lower baseline and are converging towards league norms proportionally, but the raw annual increase in three-point rate has been remarkably similar across positions. Put even more simply, centers made around thirteen times as many three point attempts in 2025 than 2013 compared to the league average of 1.6 times, but the share of shots taken from beyond the three point line increased by around 17% for both.

The next part of our analysis focuses on the expected value of shots by centers taken in different zones. We found that while expected value per shot increased across all zones during the period, shots taken from mid range consistently had the lowest expected value across years, shots from the three point line varied highly but generally were stronger than shots from the paint, and shots from the restricted area had a much higher expected value than others. We can see a significant decrease in the percentage of shots taken from mid range over the period along with the increase in three point shots discussed earlier. This signals centers are improving not only in their shot accuracy, but in choosing the best zones on the court to shoot.

Finally, we conducted the same log likelihood and linear regression tests to find differences in changes between centers and the rest of the league for each of the zones. The results revealed the change in annual log likelihood of making a shot in a zone differed between centers and the rest of the league for each zone, which is unsurprising given the large difference in playstyle. Interestingly, however, the linear regression showed no significant difference in the slope of the raw percentage increase over year between centers and the league for all zones except for mid range, where the amount of mid range shots taken by centers decreased by 0.14% more per year than the league average ($p = 0.017$). While not a large difference, this does show centers have reduced the proportion of shots taken from mid range by more than the rest of the league.

Summary and Limitations

This project set out to observe the evolution of NBA center shot location and efficiency from 2013 to 2025, We utilized data from the NBA API to observe the change in these shot patterns, and made several interesting findings. Over the period from 2013 to 2025, the three-point shot underwent a large transformation among NBA centers. The beginning of the observed period was characterized by a near negligible proportion of center shots taken from beyond the three point line, with centers playing to the position's historic strengths in the paint and restricted area with the occasional mid-range jump shot. However, year-over-year data reveals a steep increase in three-point attempt rate beginning in 2016, stabilizing in 2020 and continuing through 2025. This shift signifies a change in what it means to be a center in the modern NBA, with the once rare three-point shot steadily becoming a staple in their game.

Our comparative trend analysis produced an interesting finding. While centers adopted the three-point shot at a proportionally faster rate than the rest of the league, our OLS regression showed the absolute annual increase in three-point attempt rate was statistically indistinguishable from the league average. Given how infrequently centers made three-point attempts prior to 2016, we have a scenario where centers in 2025 were making thirteen times the three-point attempts they were in 2013, but still making them around 25% less than the league average. Centers were, however, reducing the amount of mid-range shots made by a greater raw percentage than those in the rest of the league. This reflects the trend of centers making more shots from positions with higher expected value.

There are some limitations worth discussing. The analysis relied on the NBA's position labels, which do not fully account for the fluid positional roles of modern basketball. Players listed as forward-center or power forward who have very similar roles to centers would not be considered as such in the analysis, which could bias the comparison between centers and non-centers. This analysis was also purely descriptive. We have documented changes in shot locations for centers compared to the rest of the league, but we cannot make causal claims about why this is the case. Future work could cast a wider net for players to include as centers, find which teams were early adopters of the increase in three-point attempts and compare it to their seasonal results, or determine if three-pointers were more likely to occur in different periods.

References

Gilani, Saiem. "Get NBA Stats API Shot Chart Detail — nba_shotchartdetail." *hoopR*, SportsDataverse, hoopr.sportsdataverse.org/reference/nba_shotchartdetail.html.

Yissa, Ayomide. "Welcome to the NBA-API Documentation!" *NBA-API Documentation*, 2024, nba-apidocumentation.knowledgeowl.com/help.

List of Figures and Tables

Fig 1. Center Shot Density GIF

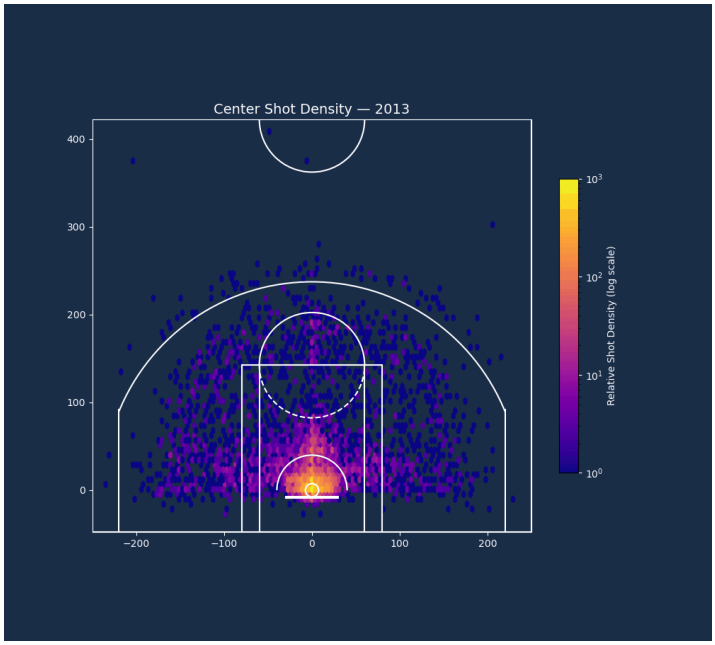

Fig 2.Center attempt rate by year

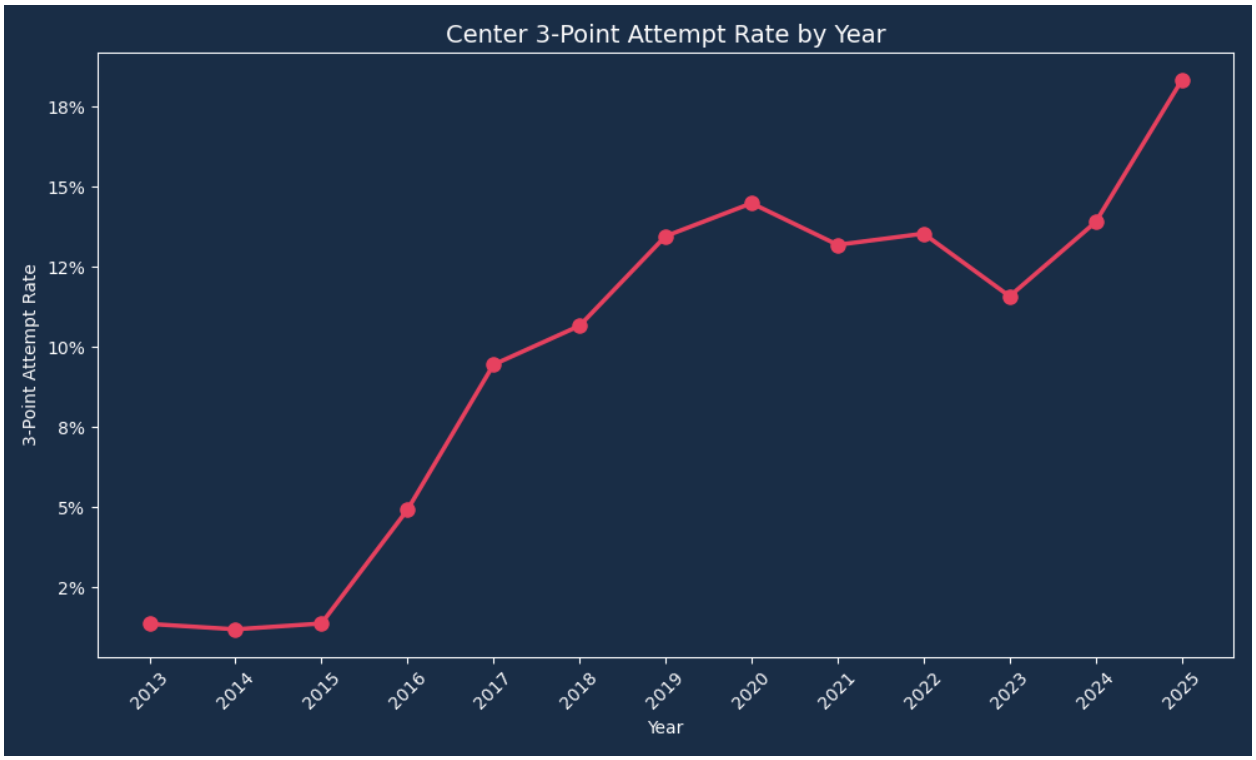

Fig 3. Shot volume by zone vs. FG% by zone

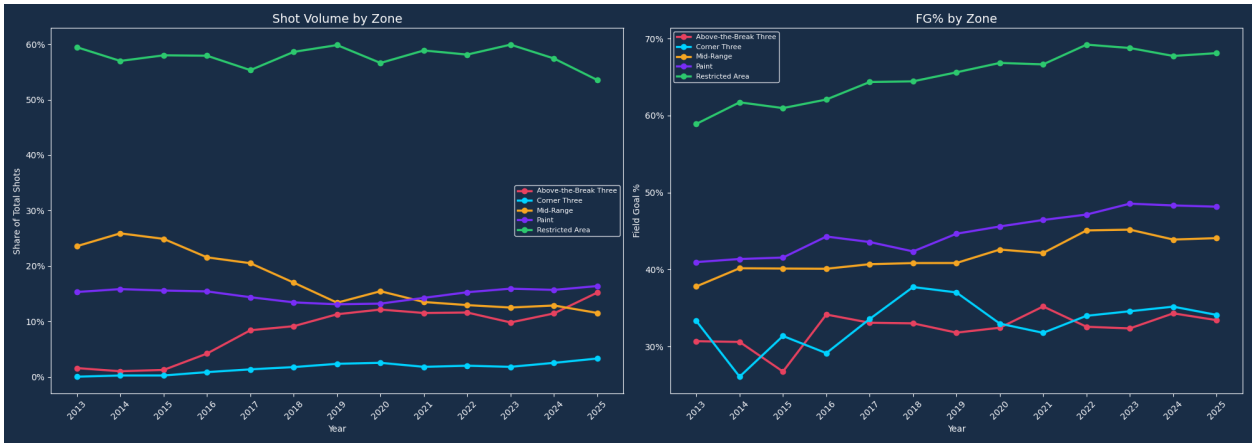

Fig 4. Expected Points By Zone

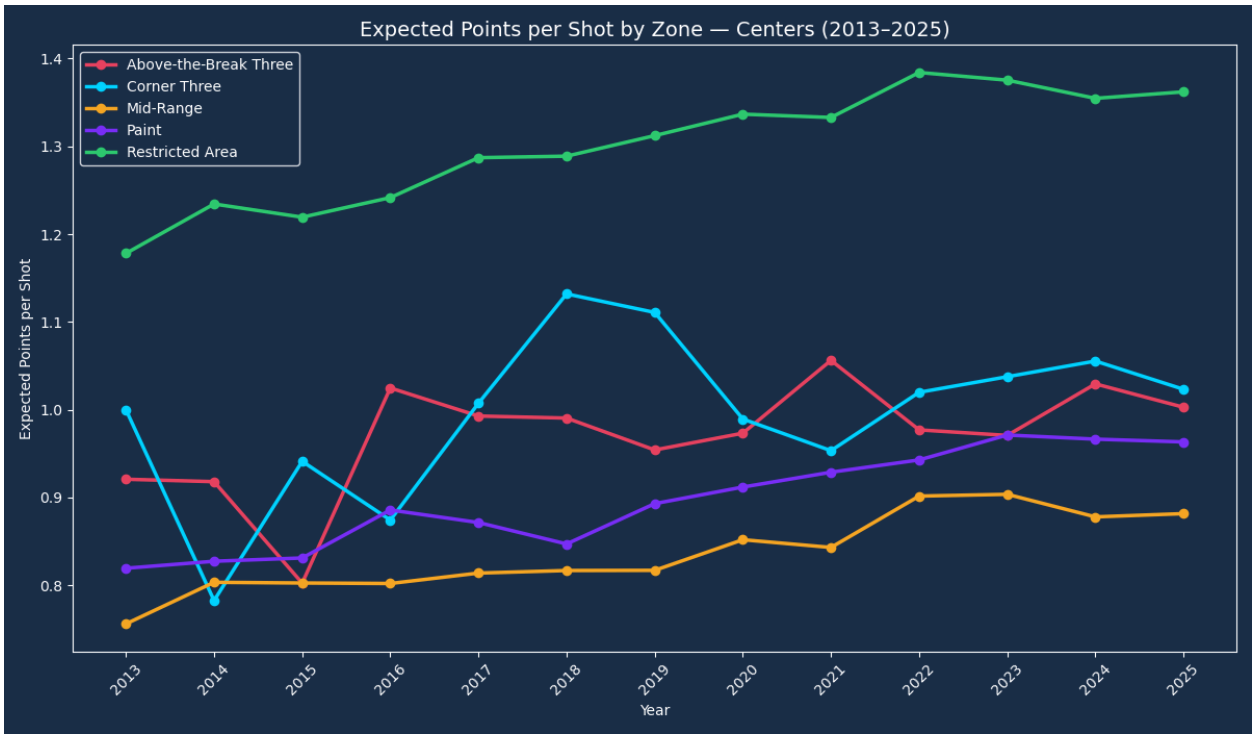
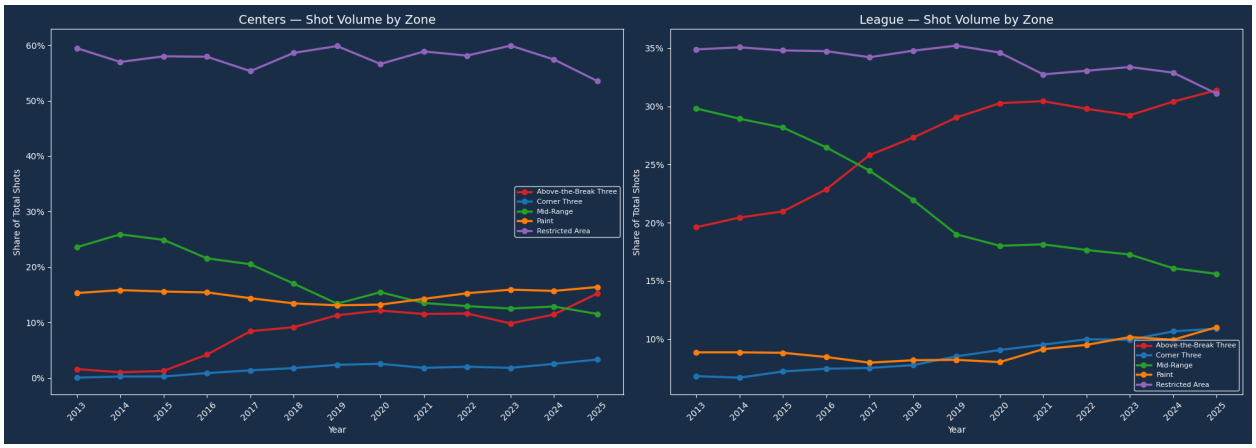
Fig 5. Volume Comparison